NPTEL Week 5

ASSIGNMENT 5 SOLUTIONS

PROGRAMMING SOLUTIONS

1. Smallest Number with Maximum Frequency

Calculates frequency of elements and prints the smallest element among those with the highest frequency.

// PYTHON

```
arr = [int(x) for x in input().split()]
freq = {}

for num in arr:
    freq[num] = freq.get(num, 0) + 1

max_freq = max(freq.values())
candidates = [num for num, count in freq.items() if count == max_freq]

print(min(candidates), end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `arr = [int(x) for x in input().split()]
freq = {}

for num in arr:
    freq[num] = freq.get(num, 0) + 1

max_freq = max(freq.values())
candidates = [num for num, count in freq.items() if count == max_freq]

print(min(candidates), end='')`);
}
```

2. First Duplicate Elements

Identifies and prints all duplicate integers in their initial repeating order. Returns -1 if none.

// PYTHON

```
arr = [int(x) for x in input().split()]
seen = set()
added = set()
result = []

for num in arr:
    if num in seen:
        if num not in added:
            result.append(num)
            added.add(num)
    else:
        seen.add(num)

if not result:
    print(-1, end='')
else:
    print(*result, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `arr = [int(x) for x in input().split()]
seen = set()
added = set()
result = []

for num in arr:
    if num in seen:
        if num not in added:
            result.append(num)
            added.add(num)
    else:
        seen.add(num)

if not result:
    print(-1, end='')
else:
    print(*result, end='')`);
}
```

3. Elements Appearing Exactly Twice

Finds elements occurring exactly two times and prints them in increasing order. Returns -1 if none.

// PYTHON

```
arr = [int(x) for x in input().split()]
freq = {}

for num in arr:
    freq[num] = freq.get(num, 0) + 1

exactly_two = [num for num, count in freq.items() if count == 2]
exactly_two.sort()

if not exactly_two:
    print(-1, end='')
else:
    print(*exactly_two, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `arr = [int(x) for x in input().split()]
freq = {}

for num in arr:
    freq[num] = freq.get(num, 0) + 1

exactly_two = [num for num, count in freq.items() if count == 2]
exactly_two.sort()

if not exactly_two:
    print(-1, end='')
else:
    print(*exactly_two, end='')`);
}
```

MCQ ANSWERS

Q01	5
Q02	It executes only when a category is not present in borrow_count.
Q03	12
Q04	max_books
Q05	MSQ —The values of "Science" and "History" are increased. —The value of "Technology" remains unchanged. —Exactly one new key is added to the dictionary.
Q06	It stores which move defeats another move.
Q07	{"Player A": 3, "Player B": 1}
Q08	The move defeated by Player A's current move.
Q09	Player A Wins
Q10	MSQ —The dictionary score is updated during program execution. —The program records one drawn round. —The statement score["Player A"] += 1 updates the value associated with the key "Player A".
Q11	Because Binary Search requires the data to be sorted.
Q12	Linear Search: 4 and Binary Search: 4
Q13	3
Q14	The Linear Search continues checking the remaining elements even after finding the target.
Q15	MSQ —Linear Search checks elements one by one from the beginning of the list. —Binary Search repeatedly divides the search space into two halves. —Both Linear Search and Binary Search return index 4 for the given input.
Q16	6
Q17	117
Q18	[121, 126, 130, 135, 140]
Q19	It eliminates half of the remaining search space after each comparison.

Q20 MSQ

- Binary Search requires the list to be sorted.
- Linear Search can be used on both sorted and unsorted lists.
- Both Linear Search and Binary Search will locate Gate 126 in this scenario.